

Decomposition of $\mathrm{HH}^2(u_q(\mathfrak{sl}_3), \mathbb{C})$: $\dim = 3 + 3 + 3 = 9$

Cartan-type

$\binom{n+1}{2} = 3$

$\xi_{K_1 K_2}$

Type I: pairwise
 $K_1^3 = K_2^3 = 1$

$\xi_{K_1}^{\text{cross}}$

Type II: $K_1 E_j$
cross relation

$\xi_{K_2}^{\text{cross}}$

Type II: $K_2 E_j$
cross relation

Positive ℓ -th power

$|\Phi^+| = 3$

$[E_1^3]$

Simple root α_1

$[E_2^3]$

Simple root α_2

$[E_{12}^3]$

Composite root $\alpha_1 + \alpha_2$

Negative ℓ -th power

$|\Phi^+| = 3$

$[F_1^3]$

Simple root α_1

$[F_2^3]$

Simple root α_2

$[F_{12}^3]$

Composite root $\alpha_1 + \alpha_2$